

KLS Gogte Institute of Technology, Belagavi

IA Test – I

Program: BE

Department: Information Science & Engineering

Course Title: Data Structures and Applications

Duration: 1 Hr.

Semester: III

Academic Year: 2024-25 (ODD SEM)

Max. Marks: 30 marks

Code: 22IS34

Date: 26/09/24

Instructions:	Write comments in the programs where ever necessary.
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Q.No	PART A (Compulsory question)	[L]	[CO]	[PO]	[M]
1a.	Define Outcome Based Education (OBE).	2		1	1
1b.	Define CO and PO.				1
1c.	Define data structure. Give examples for two data structures with their applications.				4
1d.	Explain the following file handling functions of C: fopen, feof, fscanf and fprintf along with their syntax.				4
PART B (Answer any 1 questions)					
2.	Develop a C program to read and display the information namely, Name, USN , Branch, Sem and CGPA) of 50 students using structure and display the topper of the class.	3	2	3	10
3.	Develop a C program to implement stack operations.	3	2	3	10
PART C (Answer any 1 question)					
4.	Covert the following Infix expression in to the postfix expression using tabulation method and also display the content of the stack. i) $(A+B*C-D/(E+F)/G)$ ii) $((5+4)*6)/2+(7-3)$	3	3	1	10
5.	During the promotion of students from second semester to third semester student data is captured and stored in a input file, which contains information of 'N' students (Name, USN, Branch, Semester and CGPA). Develop a C program to transfer the contents of input file to output file such that the student records are stored in an ascending order of CGPA.	4	2	3	10